

TROY UNIVERSITY
CS 256
Computer Science II
COURSE SYLLABUS
Spring 2026

PRE-REQUISITES:

CS 255 or permission from instructor

INSTRUCTOR INFORMATION:

Huu-Tu DANG, Ph.D

Dept. of Applied Mathematics and Informatics, Hanoi University of Science and Technology

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INSTRUCTOR EDUCATION:

Ph.D. in Computer Science, 2025, Toulouse Capitole University, France

M.Sc. in Mathematics and Informatics, 2021, Hanoi University of Science and Technology

B.Sc. in Mathematics and Informatics, 2020, Hanoi University of Science and Technology

CATALOG DESCRIPTION:

A continuation of the introduction to the theory and development aspects of a high level programming language. The course provides students with a foundation in problem solving, algorithm development, and program design using C++. It covers enumeration, repetition, multi-dimensional array and string manipulation, structures, advanced files input/output and an introductory to class concept in C++. Program analysis, design, development and testing are emphasized

STUDENT OUTCOMES:

- Understand Class concept in C++
- Implement solutions in object oriented programming paradigm
- Understand advanced structures such as linked lists, queue, stacks.
- Implement advanced files operation
- Analyze, develop solution and debugging problem in C++

Students will demonstrate the above through their performance on the assigned programming projects, exams and discussion Black Board exercises.

GRADING

Individual programming projects - 30%

Team programming Project - 40%

Final Exam - 30%

GRADING SCALE:

Grades will be assigned according to the following scale:

A	90 - 100
B	80 - 89
C	70 – 79
D	60 – 69
F	below 60

TEXTBOOK

Tony Gaddis, Starting out with C++, Eighth Edition, Pearson, 2015, ISBN: 978-0-13-403732-5. A C++ compiler is needed. Microsoft visual studio or C++ express are recommended for the course. You can download them free online as the instructions provided. "Computer problems" are not an acceptable excuse if you cannot turn in homework by the deadlines.

METHODS OF INSTRUCTION:

In-Classroom Lectures, Labs, Tests, Handouts/Homework, Class Participation and Reading assignments.

HONESTY AND PLAGIARISM

*Plagiarism is defined as submitting anything for credit in one course that has already been submitted for credit in another course, or copying any part of someone else's intellectual work – their ideas and/or words – published or unpublished, including that of other students, and portraying it as one's own. All students are required to read the material presented at:
<http://troy.troy.edu/writingcenter/research.html>

All material submitted for grade must be the student's own work.

Anyone found cheating and/or copying will receive an automatic 0 for that assignment or exam or dismissal from the course. This goes for the person who copies as well as the person who allows their work to be copied. Serious penalty (e.g: one lower letter grade) will be given for cheating and plagiarism and students will be required to retake a course if they get D or worse for that course.

OTHER POLICY

There will be no make up test. A missed test or exam will result in 0 points. Contact me in advance in case of emergency such as illness. An original letter addressed to me on a letterhead paper from a physician or hospital stating that you could not take the test or exam as scheduled is necessary for me to consider your case.

PROJECTS

Students are required to submit their source code file(s) as email attachments prior to the project deadline. In the event you are unable to complete a project or make it work correctly, be sure to email your latest source code to receive partial credit. Students failing to submit projects within the allotted time will receive a project grade of zero and will not be allowed to make late submissions. Exceptions will be made ONLY under extenuating circumstances and ONLY with prior approval by the instructor. To receive full credit, projects must be submitted on or before the due date.

TENTATIVE SCHEDULE

Lecture 1	Course Introduction, Pointers	Gaddis, Chapter 9
Lecture 2	Introduction to Classes	Gaddis, Chapter 13
Lecture 3	More About Classes	Gaddis, Chapter 14
Lecture 4	Inheritance, Polymorphism and Virtual functions	Gaddis, Chapter 15
Lecture 5	Individual programming project	
Lecture 6	Inheritance, Polymorphism and Virtual functions	Gaddis, Chapter 15
Lecture 7	Team project idea presentation	
Lecture 8	Individual programming project	
Lecture 9	Debugging, Exceptions	Gaddis, Chapter 16
Lecture 10	Linked List	Gaddis, Chapter 17
Lecture 11	Individual programming project	
Lecture 12	Stack & Queue	Gaddis, Chapter 18
Lecture 13	Recursion	Gaddis, Chapter 19
Lecture 14	Team project evaluation	
Lecture 15	Final Exam	

TEAM PROJECT REQUIREMENTS

1. Data must be stored in files.
2. Should use as many learned OOP techniques as possible. The program must demonstrate fundamental principles of OOP such as Abstraction, Inheritance, Polymorphism, ...
3. Must have a minimum of 4 object classes.
4. Clear task assignment! All members must participate in writing code. The program may be incomplete (if a member's assigned task is unfinished). The instructor does not evaluate based on the overall completeness of the project, but rather on the implemented functionalities and whether the overall program adheres to OOP principles.
5. The report must follow the template provided by HUST/TROY. It should be limited to 30-40 pages. The number of figures/images included should also be limited to a maximum of 15-20.

TOPIC SUGGESTIONS FOR TEAM PROJECTS

No.	Topic
1	Student Management
2	Supermarket Sales Management
3	Inventory Management (import/export)
4	Team Task Management
5	Personal Schedule Management
6	Library Book Borrowing/Returning Management
7	Human Resources Management